

YOOLKYU PARK

The University of North Carolina at Chapel Hill
Department of Statistics and Operations Research
Chapel Hill, NC

yu67667@gmail.com
[linkedin.com/in/yoolkyu-park](https://www.linkedin.com/in/yoolkyu-park)
(945) 217-0714

RESEARCH INTERESTS

Biostatistics, Computational Biology, Medical Imaging, Spatial Statistics, Topological Data Analysis

EDUCATION

The University of North Carolina at Chapel Hill

B.S. in Mathematics, Statistics and Analytics

Aug 2025 – May 2027 (Expected)

GPA: 3.61 / 4.00

The University of Texas at Dallas

Undergraduate Coursework in Mathematics and Data Science

Aug 2023 – May 2025

GPA: 3.84 / 4.00 (Transferred to UNC Chapel Hill)

PUBLICATIONS AND PREPRINTS

1. **Y. Park**, F. Wu, X. Feng, S. Yang, E. H. Wang, B. Yao, C. Moon, G. Xiao, and Q. Li(2025). Spatial Analysis for AI-segmented Histopathology Images: Methods and Implementation. Under review. ([arXiv:2512.06116](https://arxiv.org/abs/2512.06116))

RESEARCH EXPERIENCE

Department of Statistics and Operations Research, UNC Chapel Hill

Undergraduate Researcher (Senior Honors Thesis Candidate)

Aug 2026 – Present

Advisor: Dr. Zoe Huang

- Directed reading on Markov chain mixing times and cutoff phenomena.

School of Medicine, Washington University in St. Louis

Visiting Researcher

Aug 2026 – Present

Summer Research Intern (REU)

May 2026 – Aug 2026

Advisor: Dr. Fuhai Li, FuhaiLiAI Lab

- Paired scRNA-seq and scATAC-seq analysis across brain regions and drug conditions for single-cell multi-omics model development; quality control, cell-type annotation, differential expression and accessibility, and motif analysis.

Department of Mathematical Sciences, UT Dallas

Undergraduate Researcher

Aug 2024 – Feb 2026

Advisor: Dr. Qiwei Li, Qiwei Lab

- Spatial and topological quantification of cell-cell interactions in cell-level lung and oral cancer histopathology.

¹Updated September 3, 2026

PRESENTATIONS

- **Y. Park**, S. Huang, Y. Liu, K. Feng, F. Li. “Single-cell multi-omic analysis of brain data.” Poster presented at:
 - REU program poster session, WashU School of Medicine, St. Louis, MO, 2026.
- **Y. Park**, F. Wu, X. Feng, S. Yang, E. H. Wang, B. Yao, C. Moon, G. Xiao, Q. Li. “Spatial Analysis for AI-segmented Histopathology Images: Methods and Implementation.” Poster presented at:
 - East Asia Chapter, International Society for Bayesian Analysis Conference (EAC-ISBA), Daegu, South Korea, 2025. [*Best Student Poster Award*]
 - Undergraduate Research Symposium, The University of Texas at Dallas, Richardson, TX, 2025.

HONORS AND AWARDS

- Dean’s List (Fall 2025, Spring 2026), The University of North Carolina at Chapel Hill 2026
- Travel Award, Statistics Undergraduate Research Experience (SURE), SRCOS 2026
- Best Student Poster Award, EAC-ISBA Conference 2025
- Travel Award (Selected), Statistics Undergraduate Research Experience (SURE), SRCOS 2025
- Full-time Undergraduate Research Studentship (\$3,000), The University of Texas at Dallas 2024
- Dr. John Jagger Academic Excellence Scholarship, The University of Texas at Dallas 2024
- Dean’s List (Fall 2024), The University of Texas at Dallas 2024

TECHNICAL SKILLS

- **Programming & Tools:** Python, R, SQL, $\text{\LaTeX}$, Git.

SELECTED COURSEWORK

(* indicates graduate-level coursework)

Mathematics

- **MATH 549:** Graph Theory UNC Chapel Hill, Fall 2026
- **MATH 578:** Algebraic Structures UNC Chapel Hill, Spring 2026
- **MATH 550:** Topology UNC Chapel Hill, Fall 2025
- **MATH 383:** Differential Equations UNC Chapel Hill, Fall 2025
- **MATH 4334:** Numerical Analysis UT Dallas, Spring 2025
- **MATH 3351:** Advanced Calculus UT Dallas, Fall 2024
- Multivariable Calculus, Linear Algebra, Discrete Math & Combinatorics UT Dallas, 2023–2024

Statistics

- **STOR 634:** Probability Theory* UNC Chapel Hill, Fall 2026
- **STOR 435:** Probability UNC Chapel Hill, Spring 2026
- **STOR 415:** Introduction to Optimization UNC Chapel Hill, Fall 2025
- **STAT 4360:** Introduction to Statistical Learning UT Dallas, Spring 2025
- **STAT 4355:** Applied Linear Models UT Dallas, Spring 2025

Computer Science

- **COMP 683:** Computational Biology* UNC Chapel Hill, Spring 2026
- **CS 3345:** Data Structures & Algorithm Analysis UT Dallas, Spring 2025